

On improving experimental binding affinity predictions with synthetic data

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Abstract

The success of deep learning binding affinity prediction models depends critically on expanding experimental data with reliable synthetic data. We extend the Structurally Augmented IC50 Repository (SAIR) with $\approx 80K$ absolute free energy perturbation (AFEP) calculations and present two distinct data splits, SAIR-FEP and SAIR-OOD (out-of-distribution), to simulate realistic drug discovery scenarios. We compare sequence-based proteochemometric (PCM) models and state-of-the-art, structure-based deep learning models and demonstrate that PCM models can be enhanced by physics-based descriptors. While structure-based deep learning methods capture finer geometric detail, their performance is highly sensitive to the input structure. By filtering for high-confidence, co-folded complexes, we show that the performance improves predictably, whereas training on all complexes blindly does not yield performance gains. Finally, using the SAIR-OOD split, we demonstrate that simultaneous training on synthetic and experimental data improves performance on publicly available, experimental benchmarks. These results provide a clear strategy for using synthetic data to advance experimental binding affinity predictions.

1. Introduction

Predicting the binding affinity between small molecules and their protein targets is a fundamental task in drug discovery, guiding both hit identification and lead optimization. While experimental techniques provide accurate affinity measurements, they remain resource-intensive and therefore cannot evaluate the vast chemical and proteomic spaces relevant to modern therapeutic design (Cooper, 2011; Freire, 2009).

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To address these challenges, physics-based and machine-learning methods have become essential for prioritizing candidate compounds and accelerating drug discovery.

Historically, structure-based approaches such as molecular docking, scoring functions, and physics-informed simulations have served as the primary computational tools for estimating binding strength (Morris et al., 1998; Trott & Olson, 2010a; Mobley & Gilson, 2017). Although widely used, these methods often struggle with limited accuracy, complex parametrization, imperfect energy approximations, and challenges in handling protein flexibility (Cournia et al., 2017). More recently, deep learning (DL) models have gained momentum by learning nonlinear relationships directly from biochemical data. Models leveraging 3D protein–ligand structures, graph neural networks (GNNs), or molecular descriptors have demonstrated improved predictive performance (Jiménez et al., 2018; Feinberg et al., 2018; Gomes et al., 2017; Satorras et al., 2021), yet they remain heavily dependent on curated training data and carefully engineered molecular representations in order to be effective.

Structure-based deep learning models leverage 3D information from protein–ligand complexes to learn interaction patterns that correlate with binding strength. Early successes include voxelized 3D convolutional neural networks such as K_{DEEP} (Jiménez et al., 2018; Zheng et al., 2019; McNutt et al., 2021) and atomic convolutional networks (Gomes et al., 2017), which demonstrated that CNNs can capture spatially local interactions in binding pockets. GNN architectures have also achieved strong performance by encoding complexes as interaction graphs; recent examples include AEVPLIG (Valsson et al., 2025) and GEMS (Graber et al., 2025), both of which operate on a ligand graph, and (Zhou et al., 2025), which fuses protein and ligand graphs together. Hybrid physics–ML methods further integrate physics-based features with learned representations to improve generalization (Kaneriya et al., 2025).

Despite these advances, structure-based models rely heavily on large quantities of high-quality structural data and accurate binding affinity data. While open-access resources exist (such as the RCSB Protein Data Bank (Rose et al., 2016), and PDBBind (Wang et al., 2005)), these sources are often

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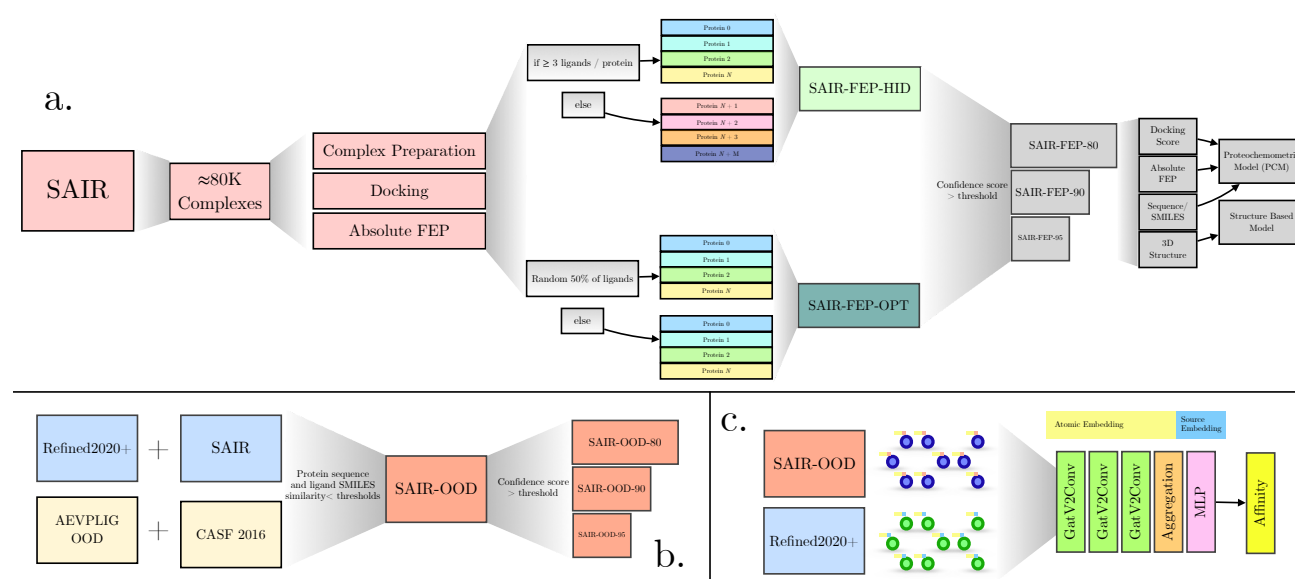


Figure 1. Overview of the work presented. a) The construction of the SAIR-FEP dataset, which includes computed docking and absolute free energies for $\approx 80K$ complexes. Hit identification (SAIR-FEP-ID) and lead optimization (SAIR-FEP-OPT) splits are done to evaluate models on different drug discovery tasks. The data is then further filtered using the co-folding confidence score and then used for training downstream models. b) The construction of the SAIR-OOD dataset, which requires filtering out complexes that have a high protein or ligand similarity with test sets evaluated (AEVPLIG-OOD and CASF 2016). This dataset is then further filtered using the co-folding confidence score. c) Depiction of the AEVPLIG model with the additional source embedding, allowing for training on synthetic and experimental complexes simultaneously.

imperfect and contain significant biases that may undermine models trained on them. For example, several analyses have demonstrated that deep networks may exploit biases in datasets such as PDBbind (Wang et al., 2005), often memorizing instead of learning true intermolecular interactions (Libouban et al., 2023). Recent advances in computational structure-prediction approaches have now made it possible to generate large amounts of accurate structural data to augment existing datasets or even to predict the bound conformation of a protein-ligand complex (Wohlwend et al., 2025). This, in theory, should enable the generation of additional high-quality structural data for training, thereby improving the performance of structure-based models. However, recent work (Hsu et al., 2025) shows that naively training on this data does not yield more performant models, and that traditional deep learning scaling laws are not observed. Thus, it remains unclear which approach is most effective for integrating synthetic data into the training sets of DL affinity models.

To overcome limitations in structural data, one can turn to SMILES and sequence-driven approaches, such as proteochemometric (PCM) models. Models such as ChemBoost (Özçelik et al., 2021) and WideDTA (Öztürk et al., 2019) treat protein sequences and SMILES strings as textual inputs, learning “chemical language” embeddings to predict affinity. More recent models integrate graph-based representations with sequence features, such as GNNSeq

(Dandibhotla et al., 2025), which combines graph neural networks with sequence-derived descriptors to achieve competitive performance without requiring protein–ligand complex structures. However, binding affinity depends critically on structural information, so structure-based models are expected to better capture the underlying physics, leading to better generalization.

Overall, the literature illustrates a diverse landscape of affinity prediction methods, each with different assumptions about data availability and molecular representation. Structure-based DL offers fine-grained geometric information, while language-based approaches provide flexibility and broad applicability. However, systematic comparisons under controlled conditions remain limited in the literature thus far. In addition, regarding structure-based approaches, it remains unclear what the best practices are for training a model on synthetic structures. This motivates our benchmarking study, which evaluates language-model-based predictors against DL and classical structure-based approaches using consistent datasets, metrics, and evaluation protocols.

In this work, we train PCM and structure-based models (AEVPLIG) on the Structurally Augmented IC50 Repository (SAIR) dataset (Lemos et al., 2025), evaluating the impact of including augmented structural data during training and comparing their performance in predicting experimentally derived binding affinity data. We first compare them directly on a subset of SAIR, called SAIR-FEP, where

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Name	N_{proteins}	N_{ligands}	$N_{\text{complexes}}$
SAIR	5149	734359	1043300
SAIR-FEP-HID _{train}	1107	48461	55083
SAIR-FEP-HID _{test}	505	22053	23519
SAIR-FEP-HID-80 _{train}	818	29765	32918
SAIR-FEP-HID-85 _{train}	609	20019	21775
SAIR-FEP-HID-90 _{train}	319	8571	9195
SAIR-FEP-HID-90 _{test}	152	3895	4015
SAIR-FEP-HID-95 _{train}	89	1435	1480
SAIR-FEP-HID-95 _{test}	43	849	863
SAIR-FEP-OPT _{train}	1223	33762	36880
SAIR-FEP-OPT _{test}	1223	34358	37506
SAIR-OOD	5089	723555	1031912
SAIR-OOD-80	3219	399054	536118
SAIR-OOD-85	2297	261812	339582
SAIR-OOD-90	1222	102144	120614
SAIR-OOD-95	285	15940	17076
Refined2020+	11130	14400	18022
Refined2020++	4822	7294	8240
AEVPLIG-OOD	189	277	291
CASF 2016	156	193	193

Table 1. Details of training and test sets used in this report. The values 80/85/90/95 indicate the confidence value (of the co-folding model) threshold applied. For example, SAIR-OOD-90 indicates that a confidence score threshold of 0.9 was applied.

we’ve computed $\approx 80\text{K}$ AFEP calculations along with various docking scores. We investigate the performance of PCM models using multiple permutations of physics-informed features to identify the most effective feature-stacking strategies for binding affinity prediction. Next, we study the performance of structure-based models on publicly available benchmarks when trained on both experimental and synthetic data simultaneously. To do so, we create another subset of SAIR called SAIR-OOD, in which all complexes that overlap with the experimental complexes were removed to avoid data leakage (Figure 1). We aim to understand model performance across various splits and to probe the underlying data to identify best practices for machine learning applied to synthetic data for binding affinity prediction.

2. Methods

2.1. Curation process of SAIR-FEP

2.1.1. THE SAIR DATASET

The SAIR dataset (Lemos et al., 2025) consists of predicted co-folded structures generated from Boltz-1x (Wohlwend et al., 2025) of 5,244,285 structures across 1,048,857 protein-ligand complexes sourced from ChEMBL (Gaulton et al., 2012; Zdrzil et al., 2024) and BindingDB (Liu et al., 2007; 2025). This structural data is also paired with experimentally derived binding data to train binding affinity

models.

2.1.2. FILTERING OF SAIR

In this work, we filter the SAIR dataset to ensure the highest quality structures are used in our analyses. For each complex, 5 conformations were generated with Boltz-1x, with multiple entries possible for certain complexes due to assay variability in IC50 measurements. We selected only the top-ranked structure for each protein-ligand complex with an aggregated confidence score (estimating reliability via pLDDT and ipTM) greater than 0.5, yielding 1,754,697 complexes. These were further filtered by retaining only biochemical assays, proteins with at least 10 measurements, and those with sufficient pIC50 variability ($\geq 2.5 \log \text{pIC50}$ range). The filtered data had 1,641 unique proteins and 100,617 entries. Next, we focus on curating 2 train/test splits that emulate realistic drug discovery scenarios. SAIR-FEP-HID (Hit Identification): This split evaluates zero-shot model generalization across unseen protein targets. We partitioned the data protein-wise: high-count proteins (≥ 3 ligands) were randomly allocated to 70% training and 30% test sets. To maximize the challenge of target generalization, all remaining low-count proteins (< 3 ligands) were exclusively assigned to the test set, ensuring the model is evaluated on rare targets. SAIR-FEP-OPT (Lead Optimization): This split evaluates model performance when proteins are shared, but the ligands are split randomly between training and test sets. This emulates a lead optimization campaign, where the target and some hit molecules are known, and the model must identify better molecules against established hits. Dataset statistics for these splits are shown in Table 1.

2.1.3. COMPLEX PREPARATION

Co-folded protein-ligand structures were initially prepared by separating the ligand and protein into their apo forms. Next, ligands were processed with RDKit (Landrum et al., 2025) to ensure correct protonation states, tautomer assignment, and to resolve stereochemistry. The binding poses generated by co-folding were reserved for Gnina scoring (McNutt et al., 2021) and AFEP calculations (Crivelli-Decker et al., 2024). Proteins were prepared to ensure proper protonation of key residues, repair missing atoms or side chains, and perform short energy minimizations to resolve steric clashes or unrealistic geometries.

2.1.4. DOCKING RESCORING USING CNN AND VINARDO SCORING FUNCTIONS

Prior to absolute binding free energy calculations, all complexes were rescored using Gnina (McNutt et al., 2021) with both a convolutional neural network (CNN) based scoring function and the Vinardo empirical scoring function (Quiroga & Villarreal, 2016). This rescoring step served as

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an intermediate triage layer to benchmark machine-learning-based and classical docking scores on the same physics-enriched structural ensemble and to enable direct comparison with subsequent AFEP-derived binding free energies. The CNN evaluates protein–ligand complexes represented as three-dimensional voxel grids to predict binding affinity and pose quality directly from learned spatial interaction patterns, while Vinardo is a physics-inspired empirical scoring function derived from AutoDock Vina (Trott & Olson, 2010b) that estimates binding affinity as a weighted sum of intermolecular interaction terms optimized for robustness and transferability.

While both CNN-based and Vinardo scoring functions provide computationally efficient estimates of binding affinity, they rely on static representations of protein-ligand interactions and limited conformational sampling. Neither approach explicitly accounts for solvent reorganization, protein flexibility, or entropic contributions to binding.

2.1.5. FREE ENERGY CALCULATIONS

For each protein-ligand complex, free energy calculations were performed using the AQFEP protocol described in (Crivelli-Decker et al., 2024). AQFEP uses an absolute free energy perturbation calculation based on the double-decoupling alchemical protocol. The double-decoupling approach is considered the “gold standard” for absolute free energy calculations by ensuring thermodynamic consistency, accurate sampling of the free-energy landscape, and broad applicability across a variety of systems. Using an absolute free energy calculation, which directly estimates the binding free energy of the given ligand-protein pair, the procedure requires much less human guidance than the more typical relative free energy calculations. Unlike relative FEP, which requires a congeneric ligand series and predefined alchemical mappings between similar compounds, AQFEP is an absolute FEP method that evaluates each ligand independently, making it directly applicable to heterogeneous libraries spanning diverse chemotypes and binding modes. This independence enables more flexible benchmarking across targets and chemistries, and facilitates systematic comparison with docking and machine-learning-based scoring methods in large-scale virtual screening workflows. Convergence was assessed independently for both the bound and unbound legs using predefined criteria, and only complexes satisfying all convergence requirements were retained for downstream analysis. Binding free energies are reported as ΔG (kcal/mol), with associated uncertainties reflecting the statistical error of the free energy estimates. In total, roughly 157,000 GPU hours (NVIDIA Tesla T4) went towards the FEP calculations.

2.2. Curation of SAIR-OOD

As mentioned in previous literature (Libouban et al., 2023), benchmarks on structure-based affinity models have revealed biases in the underlying training data, highlighting data leakage and the need for proper held-out test sets to rigorously evaluate these models. To address this, we evaluated the impact of training on two datasets: Refined2020+ (Valsson et al., 2025) (which is a specialized split of PDB-Bind2020 (Wang et al., 2005)) and SAIR (Lemos et al., 2025). To better understand our model’s generalization performance and the impact of synthetic data, we evaluated on two commonly used test sets in the literature: AEVPLIG-OOD (Valsson et al., 2025), and CASF 2016 (Su et al., 2018). However, initial analyses revealed high structural similarity between our training sets and the two test sets. As a result, we curated a second dataset, SAIR-OOD, that addresses this limitation. The curation approach is described below.

To minimize data leakage, we removed entries from the training set that had high structural or sequence similarity to the evaluation sets, as done in previous work (Valsson et al., 2025). To calculate protein similarity, we used MMSeqs2 (Steinegger & Söding, 2017), and to calculate ligand similarity, we used Tanimoto similarity between entries’ Morgan fingerprints, as included in RDKit version 2025.09.03 (Landrum et al., 2025). If the protein or the ligand had a similarity value > 0.5 , the complex was removed from the training set. With SAIR, 5 complexes were predicted per complex. However, since structure-based models require a single structure, we chose to use the complex with the lowest Vina score (Trott & Olson, 2010a); this left 1,043,300 unique structures. Filtering these complexes based on sequence and SMILES similarity left 1,031,912 structures. In addition, we further filtered this set based on the confidence scores output by Boltz-1, which are included in the SAIR dataset. We considered confidence values of 0.8, 0.85, 0.9, and 0.95, yielding 4 unique training datasets: SAIR-OOD-80/85/90/95. The number of entries, along with the unique number of proteins and ligands, is shown in Table 1.

In addition to the filtering of SAIR, we also filtered the entries of the Refined2020+ (Valsson et al., 2025) to eliminate overlap with the CASF 2016 (Su et al., 2018) test set. We refer to this training data as Refined2020++.

2.3. Training of Proteochemometric Models

We developed a Proteochemometric (PCM) modeling framework to predict the binding affinity (potency) of small molecules against multiple protein targets. This approach utilizes a supervised regression architecture that simultaneously learns from ligand chemical space and protein sequence space.

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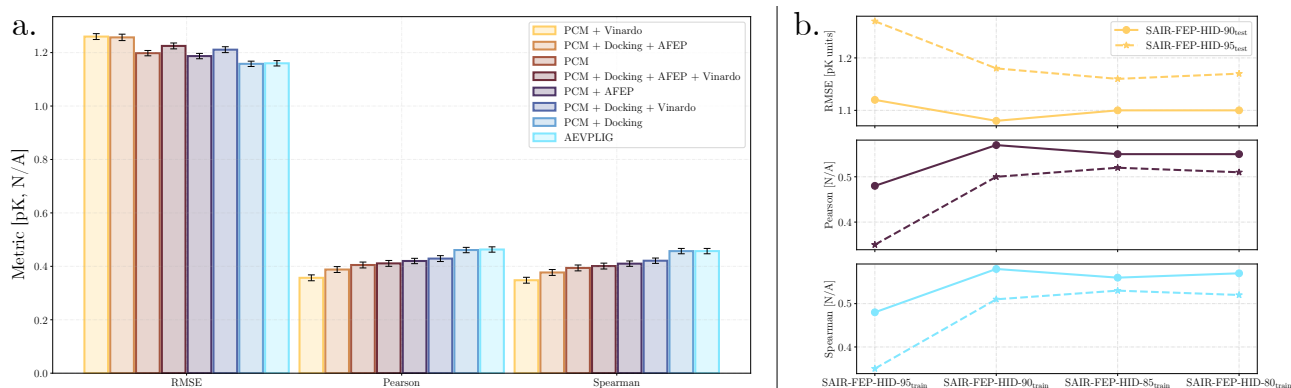


Figure 2. a) Metrics of PCM with the addition of docking and absolute free energy perturbation values as input features, as well as AEVPLIG evaluated on the SAIR-FEP-HID test set. Physics-based data enhances PCM predictions. b) Metrics tracked across the SAIR-FEP-HID-90/95_{test} sets for different training sets using AEVPLIG. Performance increases with the addition of complexes with high confidence (> 0.9), but plateaus with those of lower confidence.

2.3.1. FEATURE REPRESENTATION

Ligand Encoding: Molecules were represented as Simplified Molecular Input Line Entry System (SMILES) strings and featurized using Morgan fingerprints (2048 bits, count-based).

Protein Encoding: Protein targets were represented by their primary amino acid sequences. To capture high-dimensional biological context, we employed pre-computed embeddings from the ESM-2 (Evolutionary Scale Modeling) transformer-based language model (Lin et al., 2022). These embeddings provide a dense numerical representation of protein sequences, capturing evolutionary and structural information.

Physics-based features: To evaluate the contribution of physics-based descriptors, we systematically trained models with docking scores, including minimized affinities and deep-learning-based scoring (CNNscore, CNNAffinity) from gnina (McNutt et al., 2021) as well as free energy of binding and solvation descriptors (from AQFEP).

2.3.2. MODEL ARCHITECTURE AND OPTIMIZATION

PCM models were implemented with Pytorch (Paszke et al., 2019) using an ensemble approach that included Gaussian Mixture Network (GMN) (Bishop, 1994), Mean Variance Estimation (MVE), and Random Forest (RF). Hyperparameter selection was conducted via an automated Bayesian optimization framework using cross-validation. The search space encompassed critical parameters, including feature bit counts and network layer dimensions, while the preprocessor block remained frozen to maintain feature consistency. The optimization process was initialized with an $n = 1$ random sampling phase, followed by 9 iterations of Bayesian search directed by a surrogate probability model. The objective function was set to maximize the Spearman correlation

coefficient. Upon completion, the configuration yielding the best validation performance was selected, and the final models were refitted on the full training dataset to maximize data utility and stability.

2.4. Training of structure-based models

The structure-based model investigated in this work is a GNN called AEVPLIG (Valsson et al., 2025) which operates on molecular graphs. Briefly, the nodes and edges in the graph represent the heavy atoms of the ligand. The heavy atoms of the protein pocket are used to construct node-level embeddings but are not explicitly included in the graph passed to the GNN. An element of the embedding for heavy atom i from a particular ligand is given by

$$g_i^t = \sum_{j \in \text{Protein atoms of type } t} e^{-\eta(r_{ij}-r_s)^2} f_{\text{cutoff}}(r_{ij}) \quad (1)$$

where

$$f_{\text{cutoff}}(r_{ij}) = \begin{cases} \frac{1}{2}(\cos(\pi r_{ij}/r_{\text{cutoff}}) + 1) & \text{if } r_{ij} \leq r_{\text{cutoff}} \\ 0 & \text{if } r_{ij} > r_{\text{cutoff}}, \end{cases} \quad (2)$$

r_{ij} is the radial distance between ligand atom i and protein pocket atom j . One defines a set of values for η and r_s yielding a vector embedding for the set of protein atoms of type t . All of the different protein atoms are then concatenated to form the full input vector, $\mathbf{g}_i$. We refer the curious reader to the original manuscript for more details (Valsson et al., 2025). When training on synthetic and experimental data simultaneously, there is an additional embedding vector $\mathbf{s}$ that is concatenated with the atomic input vectors $\mathbf{g}_i$. This embedding represents the source of the data and the structure it is derived from, and can take 2 forms: synthetic or experimental.

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3. Results

3.1. Model performance on SAIR-FEP

3.1.1. INCLUDING DOCKING AND FEP RESULTS IMPROVES PREDICTIONS FROM PROTEOCHEMOMETRIC MODELS

In order to establish baseline performance of feature-based models, we first evaluated the performance of the PCM model using the SAIR-FEP-HID train/test split. As mentioned in Section 2, an array of physics-based scoring methods was included in this split to evaluate if including these metrics improves model performance. These values were appended to the existing features of the PCM model. We compare this model to a state-of-the-art structure-based model, AEVPLIG, without additional physics-based features included during training. As shown in Figure 2a, in most cases we observe improvements across all metrics when including physics-based scores as features. 3 of 4 models that including docking scores (outputs of gnina docking, which include outputs from the convolutional neural network (CNN)) all led to improvements over the baseline model. However, since some metrics from gnina depend on the output of the CNN model, it is possible that data leakage is present, potentially leading to results that do not generalize. In some cases, when combining different scoring functions together (i.e. PCM + Docking + AFEP), the metrics become worse than when using just the scoring functions independently (i.e. PCM + Docking or PCM + AFEP). This indicates that including multiple scoring functions could introduce contradictions inevitably leading to poor model performance. AEVPLIG outperformed most PCM models and was on par with the best performing PCM model which included docking scores, indicating that the non-linear mappings learned directly on structural data is enough to produce a high quality model. Interestingly, including physics-based features for docking and AFEP into node embeddings of AEVPLIG models did not result in improvements on the test set.

3.1.2. QUALITY OF CO-FOLDED STRUCTURES IMPACTS DOWNSTREAM MODEL PERFORMANCE

Our initial analysis revealed that AEVPLIG was the best-performing model on the SAIR-FEP-HID split, indicating that augmented structural data can improve downstream model performance. We now consider the effects of the quality of the underlying structural data used during model training. To start, we filter SAIR-FEP-HID with confidence scores of 0.8, 0.85, 0.9, and 0.95, each yielding its own train/test split. Confidence scores were obtained from the Boltz-1x model (see Methods). We then monitor the performance of AEVPLIG models on the 0.9 and 0.95 confidence-score test splits (called SAIR-FEP-HID-90_{test} and SAIR-FEP-HID-95_{test}, respectively) as we add more data with

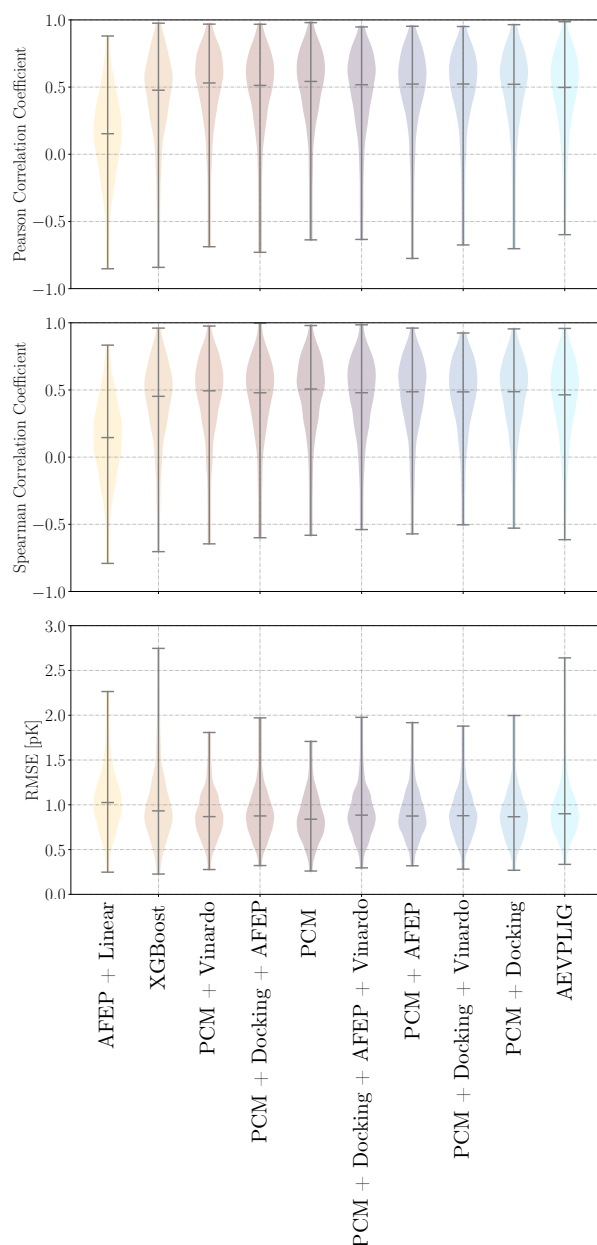


Figure 3. Comparison of Pearson/Spearman correlation coefficient and root mean squared error (RMSE) distributions across model architectures and feature integration strategies for the SAIR-FEP-OPT split for proteins with ≥ 10 ligands. Distributions represent the performance across individual protein targets. AFEP-Linear and XGBoost represent “local” models trained independently for each protein, whereas AEVPLIG and PCM-based models utilize “global” training across the entire dataset. While AEVPLIG achieves the tightest distribution overall, the PCM baseline demonstrates higher average performance than the variants augmented with additional physics-based scoring features.

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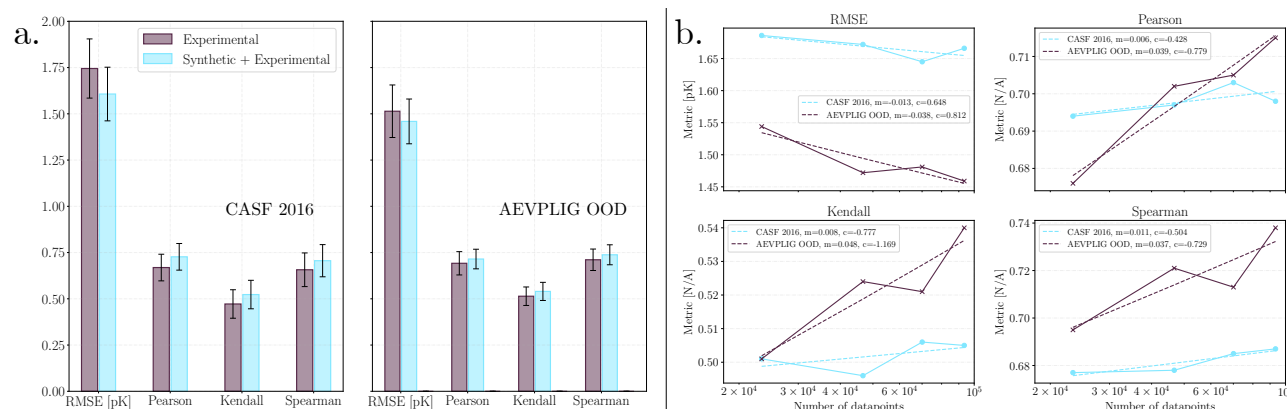


Figure 4. a) Test metrics comparing AEVPLIG models trained on experimental data alone versus models trained on synthetic and experimental data simultaneously. Consistent performance increases are seen when training on synthetic data. b) Scaling curves for various metrics versus the number of high-quality complexes (confidence score > 0.9) being trained on. Traditional scaling laws are observed, where model performance consistently improves as the dataset size increases.

lower confidence scores. As shown in Figure 2b, we observe saturation of all metrics as we include more data, which is contrary to scaling law behaviour that is typically observed in deep learning models. An interesting observation is that for SAIR-FEP-HID-90_{test} the best metrics observed were for the training set SAIR-FEP-HID-85_{train}, and for SAIR-FEP-HID-95_{test} the best metrics observed were with the training SAIR-FEP-HID-90_{train}. This suggests that a careful balance between dataset size and the quality of co-folded structures included during training has an impact on model performance.

3.1.3. MODEL PERFORMANCE IN A LEAD OPTIMIZATION SCENARIO

We now consider the SAIR-FEP-OPT data split, in which proteins are shared across the training and testing sets, but ligands are split 50/50. As discussed in Section 2, this aims to test model performance in cases where the target and hit molecules are known, and the goal is to find new molecules that surpass current hits in binding affinity. Here, we consider independent models trained for each protein and a model trained across all proteins. For the per-protein models, we use linear regression with ΔG from the AFEP calculations as the only input, serving as the physics-based baseline. In addition, we train XGBoost (Chen, 2016) models on Morgan fingerprints with 1024 bits and radius of 2, computed with RDKit (Landrum et al., 2025). For all XGBoost models, we use a max tree depth of 4, 100 models, and a learning rate of 0.1. For the all-protein model, we use PCM variants as described in Section 3.1.1 and AEVPLIG. In all cases, we compute the Pearson and Spearman correlation coefficients and root mean squared error (RMSE) for each protein that has 10 or more ligands and aggregate the coefficients across proteins. As shown in Figure 3, the physics-based linear regression model—relying

solely on AFEP-derived ΔG values without data-driven calibration—naturally yields the lowest average correlation of ≈ 0.15 with a broad spread across targets. This behavior is consistent with its sensitivity to small co-folding errors and the omission of protein reorganization and ligand entropic contributions in the fast AQFEP protocol, underscoring its value as a conservative baseline and the importance of thorough retrospective validation before prospective use. The PCM model, without additional physics based features, was the top performing model of all architectures evaluated. Interestingly, the inclusion of physics-based metrics (Docking and AFEP) into the PCM variants resulted in a performance degradation compared to the base model.

3.2. Model performance on SAIR-OOD

3.2.1. TRAINING ON SYNTHETIC DATA IMPROVES THE PERFORMANCE OF MODELS ON EXPERIMENTAL BENCHMARKS

Our previous analyses demonstrated that utilizing high-quality, structurally augmented data improves model performance on affinity prediction tasks. However, an open question in the literature is whether combining augmented data and experimentally derived structural data can improve model generalization performance. To address this, we allow the model to train on both experimental and synthetic data simultaneously and evaluate the model on two commonly used model generalization benchmarks used in the field. To accomplish this, we modified the AEVPLIG model architecture by introducing an additional embedding layer and adding an additional index to the data such that the correct embedding is selected during training and inference. This embedding was then concatenated to the node embeddings before applying the graph attention layers. A depiction of this is shown in Figure 1c. The PCM model

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requires no modification to train across synthetic and experimental data simultaneously. Although it showed good performance when training on Refined2020++ and testing on CASF 2016, we found that the addition of more data did not improve model performance. An initial baseline AEVPLIG model was trained on experimental data alone and compared to a model trained on both experimental and co-folded structures. All models were evaluated on CASF 2016 and AEVPLIG-OOD (see Section 2). As shown in Figure 4a, models trained on experimental data alone yield RMSEs from 1.5-1.7 pK units, and correlation statistics ranging from 0.51-0.71. Importantly, after the addition of synthetic data into the training set (SAIR-OD-85/90) alongside experimental (Refined2020+/Refined2020++) data, we observe performance increases on both test sets. For CASF 2016, we observe an 8% decrease in the RMSE, and a 9-11% increase in the correlation metrics compared to the baseline. For AEVPLIG-OD, we find a 4% decrease in the RMSE, and a 3-5% increase in the correlation metrics. This result suggests that including high-quality synthetic structural data improves the performance of structure-based affinity models on challenging out-of-distribution test sets.

3.2.2. NEURAL SCALING LAWS ARE RECOVERED WITH HIGH-QUALITY STRUCTURAL DATA

To understand the scaling performance of models, one typically assesses the test set performance of a model as the number of training points is increased. For structure-based deep learning models trained on synthetic protein-ligand complexes, this and previous work suggest that naively increasing the size of the dataset does not lead to performance improvements (Hsu et al., 2025). However, given that our other analyses revealed that the quality of the structure data is a key variable in model performance, we sought to replicate this pattern in our data and evaluate if adjusting the quality of structures could recover this finding. To evaluate this, we train models with different dataset sizes by changing the confidence score filter (see Table 1). We then monitor the performance of AEVPLIG on CASF 2016 and AEVPLIG-OD. In comparison to the experimental baseline, the models trained with more data also have performance gains when considering the test sets. However, similar to what is observed in Figure 2b, the metrics do not continue to improve as the number of complexes increases.

Since the addition of data that contains low-confidence structures does not yield typical model scaling behavior, we then evaluated whether improving structure quality in the training set could help to recover traditional model scaling behavior. To do so, we randomly sample 20%, 40%, 60%, and 80% of SAIR-OD-90, and train on the selected samples along with the Refined2020+ and Refined2020++ data. Next, we evaluate the performance of the AEVPLIG model on CASF 2016 and AEVPLIG-OD as a function of dataset size. Ap-

plying this quality filter allows us to roughly recover typical model scaling behavior as seen in Figure 4b. When comparing the differences in the curves between CASF 2016 and AEVPLIG-OD, the curves for AEVPLIG-OD have slopes with magnitudes 3-6 times larger than CASF 2016. This indicates that the data aggregation strategy taken to construct SAIR (Lemos et al., 2025) may have a stronger overlap with the complexes in AEVPLIG-OD than with complexes from CASF 2016. From these scaling curves, one can attempt to estimate how many complexes one would need to observe a particular metric value. As an example, to achieve a Pearson correlation coefficient of 0.9 for AEVPLIG-OD would require $\approx 32 \times 10^6$ complexes. For CASF 2016, this many complexes would only result in a Pearson correlation of 0.723, indicating once again that the data aggregation strategy taken in SAIR is currently inadequate for this test split. Taken together, these results show that by including only high-quality co-folded structures during training, we can recover traditional model scaling behavior. This paves the way for large-scale computational data augmentation efforts to build affinity prediction models that can perform effectively in extreme generalization tasks.

4. Conclusion

In this work, we investigate the role of incorporating synthetic data in improving deep-learning-based experimental binding affinity predictions for protein-ligand complexes. First, we extend the SAIR dataset by adding additional physics-based data and create two additional data splits to aid future researchers in benchmarking deep-learning affinity models. Second, we demonstrate that augmenting experimentally derived structural data with synthetic data generated by a co-folding model can improve downstream model performance on challenging out-of-distribution prediction tasks.

We initially create a core dataset called SAIR-FEP, where we perform docking and AFEP calculations for $\approx 80,000$ complexes and create two train/test splits to simulate realistic drug discovery scenarios. Next, we curate a train/test split, called SAIR-FEP-HID, which is intended to be a true out-of-domain split in which the target is not included in the training set. With this set, we train PCM models and show that including additional physics-based features (i.e., docking scores and AFEP values) improves test-set performance. We also compared the PCM models to a DL approach, AEVPLIG, which demonstrates competitive performance alongside the highest-performing physics-augmented PCM variant. However, introducing more co-folded complexes with lower confidence scores into AEVPLIG does not improve test-set metrics, contrary to the typical scaling laws observed in deep learning models.

The second train/test split of SAIR-FEP, called SAIR-FEP-

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OPT, is intended to simulate a lead-optimization scenario. The training and testing sets have the same targets, but different ligands. With this split, we compare the correlation of models trained independently for each protein to that of a model trained across all proteins. Among the global models, the PCM base model achieved the highest mean Pearson correlation, notably outperforming variants that incorporated additional physics-based features such as docking and AFEP values.

We also curate SAIR-ODD, an additional split where we remove SAIR entries that have similar complexes from the CASF 2016 and AEVPLIG-ODD test sets to prevent data leakage. We then train AEVPLIG models on synthetic and experimental data simultaneously and show that including synthetic data improves test set metrics on the experimental data. We also investigate model performance as more, lower-confidence co-folded structures are added to the training set and again observe metric saturation. To test if typical scaling laws are observed, we randomly select entries from the high-confidence complexes in increasing amounts and monitor the test metrics. Here, we find that adding more complexes improves performance, indicating typical scaling-law behavior of DL models for binding affinity prediction.

This work demonstrates that including synthetic data in various ways generally improves the predictive performance of models for experimental binding affinities. However, we show that one cannot blindly include synthetic data into their training. To achieve performance improvements, careful curation of the co-folded structures is required. Taken together, these results show that by including only high-quality co-folded structures during training, we can recover traditional model scaling behavior. This paves the way for large-scale computational data augmentation efforts to build affinity prediction models that perform effectively on extreme generalization tasks common in drug discovery.

Impact Statement

This work introduces various ways to improve model performance in predicting experimental binding affinities of small molecules to proteins by leveraging synthetic data. This work highlights the importance of synthetic structure quality when augmenting purely experimental datasets. The models developed in this work can be used for both hit identification and lead optimization in drug discovery campaigns. Ultimately, new medicines can be found using methodologies discussed in this manuscript.

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A. Results in Tabular Format

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Description	Training set	Testing set	RMSE ↓ [pK Units]	Spearman ↑	Pearson ↑
PCM + Vinardo	SAIR-FEP-HID _{train}	SAIR-FEP-HID _{test}	1.260 ± 0.011	0.348 ± 0.011	0.357 ± 0.011
PCM + Docking + AFEP	SAIR-FEP-HID _{train}	SAIR-FEP-HID _{test}	1.257 ± 0.012	0.377 ± 0.011	0.388 ± 0.011
PCM	SAIR-FEP-HID _{train}	SAIR-FEP-HID _{test}	1.198 ± 0.010	0.394 ± 0.011	0.405 ± 0.011
PCM + Docking + AFEP + Vinardo	SAIR-FEP-HID _{train}	SAIR-FEP-HID _{test}	1.225 ± 0.011	0.401 ± 0.011	0.411 ± 0.011
PCM + AFEP	SAIR-FEP-HID _{train}	SAIR-FEP-HID _{test}	1.187 ± 0.010	0.410 ± 0.010	0.420 ± 0.010
PCM + Docking + Vinardo	SAIR-FEP-HID _{train}	SAIR-FEP-HID _{test}	1.211 ± 0.011	0.421 ± 0.011	0.429 ± 0.010
PCM + Docking	SAIR-FEP-HID _{train}	SAIR-FEP-HID _{test}	1.160 ± 0.010	0.457 ± 0.010	0.461 ± 0.010
AEVPLIG	SAIR-FEP-HID _{train}	SAIR-FEP-HID _{test}	1.160 ± 0.010	0.457 ± 0.010	0.463 ± 0.010
AEVPLIG	SAIR-FEP-HID-80 _{train}	SAIR-FEP-HID-90 _{test}	1.172 ± 0.029	0.519 ± 0.028	0.525 ± 0.027
AEVPLIG	SAIR-FEP-HID-85 _{train}	SAIR-FEP-HID-90 _{test}	1.157 ± 0.027	0.524 ± 0.028	0.533 ± 0.026
AEVPLIG	SAIR-FEP-HID-90 _{train}	SAIR-FEP-HID-90 _{test}	1.181 ± 0.030	0.503 ± 0.029	0.508 ± 0.029
AEVPLIG	SAIR-FEP-HID-95 _{train}	SAIR-FEP-HID-90 _{test}	1.273 ± 0.030	0.346 ± 0.033	0.351 ± 0.031
AEVPLIG	SAIR-FEP-HID-80 _{train}	SAIR-FEP-HID-95 _{test}	1.103 ± 0.063	0.557 ± 0.064	0.566 ± 0.057
AEVPLIG	SAIR-FEP-HID-85 _{train}	SAIR-FEP-HID-95 _{test}	1.096 ± 0.059	0.550 ± 0.069	0.563 ± 0.061
AEVPLIG	SAIR-FEP-HID-90 _{train}	SAIR-FEP-HID-95 _{test}	1.085 ± 0.062	0.570 ± 0.067	0.582 ± 0.061
AEVPLIG	SAIR-FEP-HID-95 _{train}	SAIR-FEP-HID-95 _{test}	1.125 ± 0.067	0.483 ± 0.074	0.484 ± 0.068

Table 2. Metrics tracked for PCM and AEVPLIG models on various train/test splits of the SAIR-FEP-HID split. 80/85/90/95 indicates the confidence score filter applied to the co-folded data.

Test Dataset	Train Dataset	Model	RMSE ↓ [pK Units]	Pearson ↑	Kendall ↑	Spearman ↑
CASF v2016	Refined2020++	AEVPLIG	1.745 ± 0.160	0.669 ± 0.072	0.472 ± 0.077	0.657 ± 0.091
CASF v2016	Refined2020++	PCM	1.690 ± 0.160	0.690 ± 0.072	-	0.680 ± 0.080
AEVPLIG OOD	Refined2020+	AEVPLIG	1.514 ± 0.142	0.692 ± 0.063	0.514 ± 0.050	0.711 ± 0.058
AEVPLIG OOD	Refined2020+	PCM	1.800 ± 0.110	0.490 ± 0.068	-	0.520 ± 0.080
CASF v2016	SAIR-ODD-95 + Refined2020+	AEVPLIG	1.713 ± 0.149	0.684 ± 0.070	0.482 ± 0.068	0.669 ± 0.080
AEVPLIG OOD	SAIR-ODD-95 + Refined2020+	AEVPLIG	1.505 ± 0.129	0.690 ± 0.058	0.501 ± 0.053	0.697 ± 0.059
CASF v2016	SAIR-ODD-90 + Refined2020++	AEVPLIG	1.642 ± 0.154	0.708 ± 0.076	0.512 ± 0.074	0.694 ± 0.037
CASF v2016	SAIR-ODD-90 + Refined2020++	PCM	1.690 ± 0.140	0.690 ± 0.060	-	0.700 ± 0.060
AEVPLIG OOD	SAIR-ODD-90 + Refined2020+	AEVPLIG	1.471 ± 0.123	0.709 ± 0.054	0.513 ± 0.052	0.711 ± 0.056
AEVPLIG OOD	SAIR-ODD-90 + Refined2020+	PCM	1.800 ± 0.110	0.510 ± 0.080	-	0.520 ± 0.100
CASF v2016	SAIR-ODD-85 + Refined2020++	AEVPLIG	1.607 ± 0.146	0.727 ± 0.072	0.523 ± 0.077	0.706 ± 0.087
AEVPLIG OOD	SAIR-ODD-85 + Refined2020+	AEVPLIG	1.515 ± 0.131	0.683 ± 0.055	0.506 ± 0.051	0.703 ± 0.058
CASF v2016	SAIR-ODD-80 + Refined2020++	AEVPLIG	1.722 ± 0.173	0.647 ± 0.079	0.446 ± 0.074	0.618 ± 0.092
AEVPLIG OOD	SAIR-ODD-80 + Refined2020+	AEVPLIG	1.495 ± 0.149	0.698 ± 0.061	0.522 ± 0.053	0.718 ± 0.061
CASF v2016	SAIR-ODD + Refined2020++	PCM	1.840 ± 0.180	0.630 ± 0.070	-	0.640 ± 0.080
AEVPLIG OOD	SAIR-ODD + Refined2020+	AEVPLIG	1.476 ± 0.126	0.712 ± 0.059	0.527 ± 0.056	0.719 ± 0.063
AEVPLIG OOD	SAIR-ODD + Refined2020+	PCM	1.780 ± 0.120	0.580 ± 0.080	-	0.520 ± 0.100
CASF v2016	20% SAIR-ODD-90 + Refined2020++	AEVPLIG	1.686 ± 0.158	0.694 ± 0.075	0.501 ± 0.074	0.677 ± 0.088
AEVPLIG OOD	20% SAIR-ODD-90 + Refined2020+	AEVPLIG	1.544 ± 0.136	0.676 ± 0.056	0.501 ± 0.051	0.695 ± 0.058
CASF v2016	40% SAIR-ODD-90 + Refined2020++	AEVPLIG	1.672 ± 0.158	0.697 ± 0.074	0.496 ± 0.074	0.678 ± 0.087
AEVPLIG OOD	40% SAIR-ODD-90 + Refined2020+	AEVPLIG	1.472 ± 0.128	0.702 ± 0.059	0.524 ± 0.054	0.721 ± 0.061
CASF v2016	60% SAIR-ODD-90 + Refined2020++	AEVPLIG	1.645 ± 0.176	0.703 ± 0.079	0.506 ± 0.076	0.685 ± 0.091
AEVPLIG OOD	60% SAIR-ODD-90 + Refined2020+	AEVPLIG	1.481 ± 0.124	0.705 ± 0.057	0.521 ± 0.055	0.713 ± 0.062
CASF v2016	80% SAIR-ODD-90 + Refined2020++	AEVPLIG	1.666 ± 0.157	0.698 ± 0.072	0.505 ± 0.073	0.687 ± 0.085
AEVPLIG OOD	80% SAIR-ODD-90 + Refined2020+	AEVPLIG	1.459 ± 0.121	0.715 ± 0.053	0.540 ± 0.049	0.738 ± 0.054

Table 3. Metrics for AEVPLIG models trained on various splits of the SAIR-ODD data split. 80/85/90/95 indicates the confidence score filter applied to the co-folded data.